How the 1997–98 El Niño developed

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Abstract

The 1997–98 El Niño peaked at +2.28 on NOAA's Relative Oceanic Niño Index (RONI) in NDJ 1997 (+2.37 on the traditional ONI) and was documented from its rapid onset to its sudden demise by a newly completed tropical Pacific observing system [15]. We reconstruct its development from those observations, the published event studies and new reanalysis diagnostics, and grade each proposed cause by the strength of event-specific evidence. The equatorial Pacific recharged after the 1995–96 La Niña: warm water volume was positive from December 1995 and peaked in April 1997. Onto that charged state a sequence of westerly wind events from December 1996 to June 1997 — organized by two exceptionally strong Madden–Julian oscillations and amplified by twin cyclones linked to East Asian cold surges — forced downwelling Kelvin waves and pushed the edge of the warm pool east. An ocean-model decomposition attributes about half of the eastern anomaly on 1 June 1997 to that wind stress and about 40% to the initial ocean state. Coupling was established by May–June and the event grew to a December peak. Its end was set by the seasonal southward shift of the westerlies from November 1997 and came

abruptly in May 1998. Forecast models predicted warming but none predicted its strength before it was under way. The Indian Ocean Dipole of 1997 followed the Pacific warming, so it played no part in onset; whether it was forced by El Niño or arose internally remains disputed. We find no basis for attributing the event's amplitude to background warming.

Main findings

1. **Precondition (grade A−; high confidence).** Recharge after the 1995–96 La Niña: warm water volume positive from December 1995, +3.12 × 10^{14} m^3 in April 1997; about 40% of the 1 June 1997 Niño 3 anomaly comes from the initial thermal state in one ocean model [8] [14] [25].
2. **Trigger (grade A; high confidence).** Westerly wind events from December 1996 to June 1997 — calendar 1997 had the most reanalysis bursts of any year in 1979–2001 — confirmed by TAO moorings; wind stress supplies about half the 1 June anomaly and about 90% of the difference from 1996, when no El Niño developed [25] [10] [9] [18].
3. **MJO and extratropical amplification of the trigger (grade B; medium–high confidence).** Two exceptionally strong MJOs in December 1996 and February 1997 organised the bursts; East Asian cold surges and twin cyclones intensified them [29] [23] [16].
4. **Onset mechanism (grade A−; high confidence).** Kelvin-wave currents advected warm water east (central Pacific warming in late March – early April 1997), and the March event's own SST response drew in further westerlies [29] [27] [28].
5. **Amplifier (grade B; high confidence).** Bjerknes feedback from May–June 1997, with zonal advection important in the central Pacific [8] [18] [54].
6. **Termination (grade A; high confidence).** Seasonal southward shift of the westerlies from November 1997, eastward-propagating thermocline shoaling and basin discharge; abrupt surface cooling in May 1998, with an MJO implicated in its timing [57] [58] [59] [56].
7. **Indian Ocean (not a cause of onset; origin disputed; grade B).** The 1997 dipole began after Pacific coupling. One line of work traces it to El Niño's effect on the Indian Ocean winds; another argues it was mainly internal Indian Ocean dynamics [51] [50] [47].
8. **Predictability.** No model forecast even half the peak amplitude at 6–11 months' lead; the rapid succession of bursts in early 1997 was a rare draw from the atmosphere's noise [62] [60] [40].

1. Introduction

By 1997 the tropical Pacific was far better observed than in 1982. The Tropical Ocean–Global Atmosphere (TOGA) programme had built the TAO array of moored buoys, an island tide-gauge network and satellite missions for surface winds, SST and sea level [3]; TOPEX/Poseidon altimetry covered 1992 onward [19]; and 1997–98 was the first major event predicted in real time by numerous forecasting groups [62]. The 1997–98 El Niño nevertheless surprised forecasters in both its speed and its strength, and its abrupt end in May 1998 was also unexpected [15] [18] [60].

The event therefore offers a test that 1982–83 could not: whether the processes proposed for El Niño onset — recharge of equatorial heat, wind-forced Kelvin waves, intraseasonal westerly bursts and coupled feedback — can be separated and quantified for a single event when the ocean is observed directly. A lively debate followed over whether the intense westerly wind events or the slower oceanic preconditioning mattered more [25]. This paper reconstructs the sequence month by month and grades each factor, using the same framework, grades and index (RONI) as Paper 1.

2. Definitions, data and methods

Index. RONI (the three-month mean of relative Niño 3.4, i.e. Niño 3.4 minus the 20°N–20°S tropical-mean anomaly, rescaled) is the primary index [1] [2]; the traditional ONI is shown alongside. For this event the two agree closely (peak +2.28 vs +2.37), so the classification as a very strong event does not depend on the index choice.

Observations. TAO/TRITON moorings provide daily surface winds, SST and thermocline depth along the equator; TOPEX/Poseidon sea level gives Kelvin and Rossby wave signals; NOAA CPC and PMEL indices give monthly SST, SOI, 850 hPa trades, OLR, upper-300 m heat content (180°–100°W, 1981–2010 base) and warm water volume (WWV) [8] [3] [14]. SST indices are from OISST v2.1 [5] and ERSSTv5 [4].

Reanalysis diagnostics. To compare with 1982–83 on identical definitions, the westerly-burst and MJO diagnostics of Paper 1 were re-run for July 1995 – December 1998: daily NCEP/NCAR Reanalysis 1 winds and interpolated OLR, anomalies against a 1979–2000 mean plus three annual harmonics; a burst is a 5°S–5°N 1000 hPa anomaly of at least 5 m s^{-1} for at least 3 days over at least 10° of longitude within 120°E–150°W. MJO activity is described with the Wheeler–Hendon RMM index and OLR-based measures, whose amplitudes can differ for the same event [6] [7]. Daily TAO/TRITON winds and 20 °C isotherm depth on the equator give an independent, directly observed check on the reanalysis bursts and trace the Kelvin waves; the definitions (westerly ≥ 5 m s^{-1} on ≥ 2 days at 147°E–180°; intraseasonal isotherm peaks ≥ 8 m) were fixed before the results were examined [9] [10].

Evidence grades. A: event-specific observation plus a model or budget study isolating the effect. B: event-specific observation with a plausible mechanism and supporting studies. C: correlation, composite or general theory only. D: speculative, disputed or untested for this event. The grade describes the link to 1997–98, not the general validity of a mechanism.

Limits of this review. For several central studies only the abstract could be read (the reference list records what was read for each); conclusions that depend on details beyond an abstract are marked as such.

3. Pre-event conditions, mid-1995 – early 1997

The event grew out of the 1995–96 La Niña, which reached RONI −1.02 in OND 1995 [8]. As the trades relaxed after that La Niña the equatorial warm-water reservoir refilled: WWV turned positive in December 1995 and stayed positive throughout 1996 (+0.26 to +0.85 × 10^{14} m^3), while heat content east of the Date Line hovered near zero (−0.47 to +0.17 °C) [8]. This is the recharge state that recharge-oscillator theory and the 1980–99 WWV record identify as the precondition for El Niño [13] [14] [12].

At the surface, 1996 stayed ENSO-neutral and slightly cool (RONI −0.20 to −0.37 from MAM 1996 to DJF 1997), with the far eastern Pacific cold (Niño 1+2 −0.7 to −1.6 °C) [8]. In autumn 1996 the western Pacific was warm and the warm pool extended east, a pattern also present before the aborted event of 1989–90 and before 1982 [29]. Between November 1996 and January 1997 the eastern Pacific had cold equatorial SST and high sea-level pressure, while off-equatorial warm SST and anomalous cyclones in the far west accompanied a switch of equatorial wind anomalies from easterly to westerly [17].

Recharge alone did not determine the outcome. An adjoint analysis with an ocean general circulation model attributes about 40% of the 1 June 1997 Niño 3 anomaly, relative to an average year, to the initial thermal state on 1 December 1996 — concentrated in the deep layer of warm water along the equator in the western Pacific, "in agreement with a recharge mechanism" — and about 50% to the wind. Compared with 1996, a year with a weaker but real build-up and no El Niño, about 30% of the difference comes from the stronger build-up and about 90% from the strong westerly wind events that 1995–96 lacked (the shares exceed 100% because other terms were negative) [25].

4. Development, December 1996 – December 1997

Westerly wind events, December 1996 – June 1997. A series of westerly wind bursts from December 1996 to June 1997, notably in March 1997, was instrumental in setting up the event [18]. In the adjoint analysis three wind-stress peaks — early March, late March and early April 1997 — contribute about 0.6, 0.3 and 0.5 °C to the modelled 1 June Niño 3

anomaly, and a strong December 1996 event about 0.4 °C over a negative baseline; each generated downwelling Kelvin waves that deepened the eastern warm layer two to three months later [25]. The March event was the strongest of the winter in stress (up to 0.2 N m^{-2}), fetch (more than 30° of longitude) and duration (about a month, mid-February to mid-March) [27]. Satellite (SSM/I) winds show westerly anomalies almost every month from September 1996 to March 1997, with the late-December and late-February episodes particularly intense and long-lived; they were embedded in the active phase of the MJO but greatly enhanced over the western Pacific by equatorial twin cyclones, themselves induced by northerly cold surges from East Asia and the western North Pacific [23]. A best-track study of the same period finds twin tropical cyclogenesis tied to MJO westerly bursts in mid-October 1996 over the Indian Ocean and in late May 1997 over the western Pacific [24]. Two exceptionally strong MJOs, in December 1996 and February 1997, first cooled the western Pacific through cloud shading and evaporation, with off-equatorial upwelling adding to the cooling in February [29]. On Paper 1's definitions the reanalysis contains 10 bursts between December 1996 and June 1997 and 16 in calendar 1997 — the most of any year in 1979–2001 (mean 4.2 a year; 1982 had 13, while 1995 and 1996 had 0 and 3). The strongest were on 20–31 December 1996 (+11.8 m s^{-1} at 137.5°E), 3–19 March 1997 (+10.3 m s^{-1} at 150°E) and 10–21 April 1997 (+10.8 m s^{-1} at 155°E). TAO moorings observed the same episodes directly, with daily westerlies of up to 8.7 m s^{-1} at 156°E in late December, 9.9 m s^{-1} at 147°E in late February – early March and 11.0 m s^{-1} at 165°E in mid-April [10] [9].

Kelvin waves and the eastward jump of the warm pool. In TAO records from October 1996 to December 1997, intraseasonal variability east of the Date Line was dominated by first-baroclinic Kelvin waves travelling east at 2.4–2.7 m s^{-1} [20], consistent with the 2.4 m s^{-1} found for the decade before [31]. The onset was linked to such waves forced by intraseasonal oscillations originating over the Indian Ocean [16]. The TAO thermocline record traces two of the resulting downwelling Kelvin waves: an intraseasonal deepening at 165°E on 02 January reached 110°W on 23 February (about 2.1 m s^{-1}), and one at 165°E on 16 March reached 110°W on 23 April (about 2.88 m s^{-1}) [9]. The first wave's sea-level signal reached the South American coast in February, and a second, larger group arrived in April–May, greatly depressing the eastern thermocline [23]. In late March and early April 1997 the Kelvin wave forced by the February MJO warmed the central Pacific, mainly by zonal temperature advection; afterwards winds, currents and SST gradients did not return to their earlier state [29]. In ocean-model experiments the March event displaced the eastern edge of the warm pool about 2000 km in a month, cooled the far west by about 0.8 °C and warmed the Kelvin-wave path by about 0.5 °C [26] [27]; a nonlinear interaction with the salinity and temperature fronts at the warm-pool edge tripled the surface currents relative to a linear response [28]. In an atmospheric model forced with those SST changes, two responses favoured rapid growth: weaker trades in the central-eastern Pacific, and an

eastward shift of convection that promoted further frequent and intense westerly events in the following months [27].

Warming in two places. SST warmed first near the Date Line in January–February 1997 and in the far east from March (Niño 1+2 from −0.45 °C in February to +0.80 °C in April) [17] [8]. The eastward-moving and westward-spreading warm anomalies merged; once the central and eastern warm regions joined, coupling between SST and winds became fully effective [17] [18]. WWV peaked at $+3.12 \times 10^{14}$ m³ in April 1997 [8].

Coupled growth, May – December 1997. RONI rose from +0.83 (AMJ) to +1.20 (MJJ), with SOI −1.3 and −1.4 and central-Pacific OLR near −26 W m^{-2} [8]. Growth continued to SON (+2.22) and the peak in NDJ (+2.28); in October the central-Pacific trade index reached −4.1 and OLR −49.8 W m^{-2} [8]. Much of the western warm water was carried east by anomalous surface currents [18]. Interannual heat budgets from TAO moorings show zonal advection important everywhere and most significant in the central Pacific, vertical heat flux especially important in the east, and surface heat fluxes generally damping the anomalies [54]. Eastern equatorial SST set a new monthly record every month from June to December 1997, the cold tongue failed to form, and the strongest westerlies and deep convection occurred only over water warmer than about 29 °C [15]; by December cold-tongue anomalies averaged nearly 4 °C, with maxima above 5 °C, and the eastern thermocline was more than 90 m deeper than normal [15] [16].

5. Termination and transition, November 1997 – mid-1998

Discharge began before the surface peak. WWV fell to $+0.66 \times 10^{14}$ m³ in November and turned negative in December 1997 [8]. From November, near-Date-Line westerly anomalies shifted south of the equator, a shift reproduced as the atmosphere's deterministic response to the observed SST field; the oceanic response was thermocline shoaling that began in the central Pacific and propagated east [58] [57]. The same seasonal migration of the warmest water appears as a general mechanism of El Niño termination in observations and coupled models [55] [59].

Eastern SST stayed warm from December 1997 to May 1998 despite strong subsurface cooling [57]. Upwelling signals arrived from several directions: Kelvin waves generated by easterlies and boundary reflection from the west, Rossby waves from westerly forcing in the east, and the general uplift of the thermocline as the basin discharged [18] [16] [19]. In the west, off-equatorial pressure turned high and easterly anomalies developed over the far western Pacific [17]. When the trades returned to near-normal strength in the eastern Pacific in mid-May 1998, SST at 0°, 125°W fell 8 °C in 30 days, more than ten times the normal rate of cooling for the season [15], and eastern equatorial SST fell by more than 4 °C within two weeks [57]; the cooling began between 125°W and 170°W [16]. The cause of

the abrupt return of the trades was unknown when McPhaden wrote, and no forecast anticipated it [15]; Takayabu et al. relate the termination to an MJO [56], and an MJO was indeed active from late April to the end of May 1998 [10]. La Niña followed (RONI JJA 1998 −1.05), with the largest equatorial phytoplankton bloom then observed [8] [22].

6. Interactions with the broader climate system

Intraseasonal variability and bursts. The MJO was unusually active through the onset: the PSL OLR MJO index stayed at or above 1 almost continuously from late October 1996 to mid-June 1997, intraseasonal OLR variance over the Indian Ocean and western Pacific in October 1996 – March 1997 was the largest of 22 seasons in 1979–2001, and the bursts of 25 December, 13 March and 28 March occurred at index amplitudes of 2.7–3.3 [10]. Indices disagree for some bursts: on 13 April the OLR-based index was 2.0 but a circulation-weighted index only 0.9, as expected when indices weight convection and winds differently [7]. In a wave model forced with observed winds, MJO-forced downwelling Kelvin waves reached 15–20 cm in sea level, an order of magnitude larger than the waves reflected from the western boundary [16]. Yet an active MJO in January–April 1996 produced no El Niño, and the late-1996 bursts intensified to over 9 m s^{-1} in the Pacific while the MJO's own westerlies had fallen below 2 m s^{-1} over the Maritime Continent [23]. Across many events, MJO activity in the western Pacific in late boreal spring leads El Niño the following autumn [33], and stronger MJO Kelvin-wave forcing preceded larger eastern anomalies by 6–12 months in 1980–99 [32]. Yet waveguide warming follows MJO events only when they contain a westerly wind event, and an MJO does not make a burst more likely [35]. For 1997 the two meet: the bursts that mattered were embedded in strong MJOs [23] [29]. An intermediate coupled model initialised on 1 January 1997 produced a weak warm event on its own, and one about 50% stronger when the rectified effect of MJO-like winds was imposed [30].

Are bursts a cause or a response? Bursts become more likely as the warm pool extends east, and treating them as partly SST-dependent roughly doubles modelled event amplitude [37] [38]; statistics also show bursts migrating east ahead of the El Niño peak [36]. In 1997 the warm pool had extended east by autumn 1996 [29], so some burst activity may already reflect the developing state. This does not remove the trigger role: the decomposition finds the December, March and April events to be the main difference from 1996 [25], and the impact of a burst depends on its timing relative to the oscillation and on the background state [39].

Extratropical influence. The best-documented pathway is synoptic: East Asian cold surges induced the twin circulations that intensified the December and February bursts [23]; best-track twin cyclones, in contrast, appear in October 1996 and late May 1997 [24], so the two studies differ in what they count as a twin-cyclone event. The Pacific meridional mode and

seasonal footprinting precede many El Niños [42] [43] [44], but no study read here isolates them for 1997 (grade C).

Decadal background. The PDO was strongly positive through 1997, and the shallow meridional overturning circulation was at a low during 1992–98 before rebounding [8] [45] [46]. These describe the background; their contribution to this event is not quantified (grade C).

Indian Ocean. Cool SST appeared in the eastern Indian Ocean along and south of the equator from strong upwelling along the equator and Sumatra, and warm anomalies developed in the west from July 1997, peaking in February 1998 [49]; the CPC dipole index rose from +0.55 °C in July to +1.39 °C in November [8]. One analysis attributes the basin's SST changes largely to ENSO's effect on the atmosphere — the reversal of the Walker circulation and displaced south-east trades [50]; Webster et al., by contrast, concluded that the 1997–98 anomalies "may primarily be an expression of internal dynamics" of the Indian Ocean: the Indian Ocean rim climate differed from the usual El Niño pattern, the ENSO correlation with the along-equator SST gradient is not significant (+0.19), and of 16 years with a reversed gradient in 1950–98 only three were El Niño years [47]. Saji et al. defined a dipole mode of the tropical Indian Ocean the same year (title only read) [48]. A reanalysis and altimetry study traces the triggering to El Niño itself: divergent easterly anomalies over the equatorial Indian Ocean in summer 1997 weakened the western westerlies and strengthened the eastern easterlies, producing the east–west SST contrast by autumn through evaporation and upwelling, which ocean Rossby and Kelvin waves then reinforced [51]. Whatever its origin, the dipole developed after Pacific coupling (western warming from June, eastern cooling from July 1997) [47], so it cannot have contributed to onset; coupled forecast experiments suggest an extreme IOD can enhance El Niño [52], but this was not tested here for 1997.

Tropical Atlantic. The tropical North Atlantic was near or slightly below normal in early 1997 (−0.37 °C in March) and warm in 1998 (+0.51 °C in May) [8]. Spring cooling in the north tropical Atlantic has been proposed as an El Niño trigger in general [53]; for 1997 this is untested (grade C/D), and the 1998 warming is the usual lagged response.

Volcanic and external forcing. The Smithsonian Global Volcanism Program catalogue lists no eruption of VEI ≥ 4 anywhere in 1995–97; the most recent tropical VEI 4 eruption was Rabaul (Papua New Guinea, 4°S) in September 1994, more than two years before the December 1996 burst [11]. Volcanic forcing is therefore not a candidate for this event.

Background warming. The tropical-mean adjustment in RONI changes the peak by only 0.09 °C (RONI +2.28, ONI +2.37) [2] [8]. Contemporary debate concerned whether the post-1976 run of warm events was unusual [63]; projections of more frequent extreme El

Niños under greenhouse warming [64] do not attribute this event. No event-specific attribution is made here.

7. Causal synthesis and competing explanations

The most plausible sequence is: (1) recharge after the 1995–96 La Niña created a deep warm-water reservoir by late 1996; (2) a run of strong, MJO-organised and extratropically amplified westerly wind events from December 1996 to June 1997 forced downwelling Kelvin waves and moved the warm pool's edge east; (3) the resulting central-Pacific warming in March–April and coastal warming from March merged by May–June, when the Bjerknes feedback took over; (4) the seasonal southward shift of the westerlies from November and basin discharge ended the event abruptly in May 1998.

The main competing view weights the slow ocean more heavily: with recharge and western-boundary reflection already favourable [18] [19], an El Niño was likely regardless of the bursts. The available quantitative answer splits the difference — roughly 40% initial state and 50% wind stress at 1 June [25] — but comes from one ocean model forced with one wind product. A noise-driven ensemble started at the end of 1996 favours a moderate event over the observed extreme [40], which supports the view that the bursts set the amplitude rather than the occurrence. Bergman et al. reach a similar conclusion for the MJO: relevant to timing and initial growth rather than responsible for the event itself [29].

Unresolved: how much of the burst activity was itself a response to the eastward-extended warm pool; whether the Indian Ocean dipole was forced or internal, and whether it fed back on the peak; and the role of extratropical modes beyond the documented cold surges.

8. What was forecast at the time

This was the first very strong El Niño predicted in real time by many groups. Most models forecast some warming one to two seasons before the boreal-spring onset, but none predicted its strength until it was already becoming very strong in late spring; the best forecasts of Niño 3.4 before positive anomalies appeared were about +1 °C against an observed +2.5 to +3 °C [60]. Against a climatology-and-persistence baseline, no model anticipated even half the peak amplitude at 6–11 months' lead, and none was skilful for both onset and decay at 6–8 months [62]. Once the event had developed, more models captured the late-1997 peak and the 1998 reversal [60]. The two assessments differ in emphasis: NCEP's own review finds that its coupled model's spring-1997 forecasts, "while somewhat too weak", clearly suggested a developing strong El Niño, and that comprehensive coupled models (NCEP, COLA, ECMWF) outperformed simple dynamical and most statistical models [61]; the comparison with a persistence baseline finds no model skilful at 0–8 months [62]. Both agree that the rapid spring onset was not foreseen in its full strength. The rapid

succession of bursts in early 1997 was a relatively rare event, which limits predictability [40] [25].

9. Uncertainties

- The 40/50% decomposition comes from one ocean model and wind product [25].
- Saji et al. (1999) and Takayabu et al. (1999) were read only as bibliographic records; several other studies only as abstracts (see the reference list).
- Reanalysis burst and MJO diagnostics depend on index and definition; RMM and OLR-based indices can disagree for the same event [7].
- Off-equatorial and sub-thermocline conditions before Argo are less certain than the TAO line itself.

10. Conclusion

The 1997–98 El Niño developed from a recharged Pacific, but its speed and size came from a rare run of westerly wind events in December 1996 – June 1997, organised by strong MJOs and intensified by twin cyclones tied to East Asian cold surges. Those events forced Kelvin waves and an eastward jump of the warm pool that joined central and eastern warming by May–June, after which coupled feedback carried the event to a December peak. Its demise, set in motion by the seasonal southward shift of the westerlies, was as abrupt as its onset. Compared with 1982–83, the trigger is far better documented: the ocean was observed directly, and a quantitative separation of precondition and wind forcing exists for this event.

Evidence base: scope, index and data

Primary index: NOAA's Relative Oceanic Niño Index (RONI), official since 1 February 2026 [1] [2]. On RONI the 1997–98 peak is +2.28 (NDJ 1997); on the traditional ONI it is +2.37. The two agree within 0.1 °C for this event, unlike 1982–83 (RONI +2.40, ONI +2.14).

Observing system in 1997. The first strong El Niño observed from start to finish by the TOGA observing system — the TAO array of moored buoys, drifting buoys, island tide gauges, ship XBT lines — and by satellite winds, SST and TOPEX/Poseidon sea level (from 1992) [3] [15] [18] [19]. Onset, growth and termination are therefore documented daily along the equator, which allows heat-budget and wave analyses that were impossible for 1982. Argo did not yet exist; the subsurface off the equator and outside the array remains less certain.

Monthly indices, Jul 1995 – Dec 1998

Values from NOAA CPC, PMEL and NCEI files as held in the site's data archive; read from the source files, not retyped.

Month	Season	RONI	ONI	rel. Niño 3.4 (OISST)	Niño 3.4 (OISST)	Niño 1+2 (OISST)	SOI	W Pac trades	C Pac trades	C Pac OLR	H c 1 1
1995-07	JJA 1995	-0.41	-0.3	-0.13	-0.21	-0.5	0.4	0.7	0.3	8.2	-
1995-08	JAS 1995	-0.57	-0.55	-0.41	-0.57	-0.71	0.3	1.1	0.6	11.09	-
1995-09	ASO 1995	-0.76	-0.81	-0.48	-0.66	-0.35	0.3	0.6	0.4	5.47	-
1995-10	SON 1995	-0.93	-0.97	-0.76	-0.94	-0.82	0.0	0.8	0.3	13.87	-
1995-11	OND 1995	-1.02	-1.05	-0.92	-1.01	-0.1	0.0	1.2	0.5	11.13	-
1995-12	NDJ 1995	-0.99	-1.01	-0.82	-0.91	-0.86	-0.5	0.4	-0.6	16.36	-
1996-01	DJF 1996	-0.87	-0.88	-0.67	-0.83	-0.69	1.0	0.5	0.8	24.42	-
1996-02	JFM 1996	-0.75	-0.7	-0.68	-0.77	-0.49	0.3	0.6	1.2	26.52	-
1996-03	FMA 1996	-0.58	-0.52	-0.57	-0.66	-0.45	1.1	0.7	0.9	18.02	(
1996-04	MAM 1996	-0.37	-0.36	-0.27	-0.49	-1.64	0.8	1.1	0.6	1.06	(
1996-05	AMJ 1996	-0.25	-0.25	-0.24	-0.51	-1.34	0.3	0.1	0.6	6.8	-
1996-06	MJJ 1996	-0.24	-0.24	-0.31	-0.48	-1.31	1.2	0.8	0.6	7.89	(
1996-07	JJA 1996	-0.2	-0.22	-0.16	-0.37	-1.34	0.7	0.5	0.5	5.57	-
1996-08	JAS 1996	-0.2	-0.27	-0.08	-0.31	-1.11	0.7	1.0	-0.1	5.8	-
1996-09	ASO 1996	-0.2	-0.3	-0.12	-0.35	-1.08	0.6	0.3	0.6	6.97	-
1996-10	SON 1996	-0.25	-0.33	-0.32	-0.55	-0.73	0.6	0.9	0.4	9.57	-
1996-11	OND 1996	-0.3	-0.39	-0.14	-0.43	-1.29	-0.1	0.5	0.4	9.16	-
1996-12	NDJ 1996	-0.29	-0.41	-0.29	-0.6	-1.16	0.9	-0.7	0.7	8.63	-
1997-01	DJF 1997	-0.21	-0.41	-0.28	-0.61	-0.93	0.5	-0.7	-0.2	12.21	(
1997-02	JFM 1997	-0.03	-0.26	-0.03	-0.39	-0.45	1.7	0.4	0.9	13.12	1
1997-03	FMA 1997	0.17	-0.03	-0.06	-0.36	0.23	-0.4	-3.1	-0.6	-11.65	1
1997-04	MAM 1997	0.47	0.3	0.25	-0.11	0.8	-0.6	-2.0	-0.7	-6.54	2
1997-05	AMJ 1997	0.83	0.71	0.71	0.4	1.83	-1.3	-1.8	-0.7	-25.7	2
1997-06	MJJ 1997	1.2	1.13	1.05	0.81	2.79	-1.4	-1.6	-2.6	-26.84	2
1997-07	JJA 1997	1.5	1.48	1.35	1.27	3.59	-0.8	-0.7	-0.6	-28.74	1
1997-08	JAS 1997	1.84	1.79	1.74	1.68	3.71	-1.4	-2.5	-2.0	-31.63	1
1997-09	ASO 1997	2.07	2.04	1.9	1.84	3.63	-1.4	-2.1	-2.8	-32.45	2
1997-10	SON 1997	2.22	2.24	1.99	1.96	3.57	-1.5	-2.5	-4.1	-49.77	2

Month	Season	RONI	ONI	rel. Niño 3.4 (OISST)	Niño 3.4 (OISST)	Niño 1+2 (OISST)	SOI	W Pac trades	C Pac trades	C Pac OLR	H c 1 1
1997-11	OND 1997	2.25	2.34	2.07	2.11	3.73	-1.2	-0.0	-3.1	-31.2	2
1997-12	NDJ 1997	2.28	2.37	2.02	2.1	3.73	-1.0	-0.4	-2.2	-36.5	1
1998-01	DJF 1998	2.13	2.22	2.0	2.03	2.88	-2.7	-0.1	-2.8	-38.38	(
1998-02	JFM 1998	1.82	1.91	1.59	1.74	1.91	-2.0	-0.1	-2.7	-31.61	-
1998-03	FMA 1998	1.22	1.4	1.1	1.18	1.91	-2.4	-0.4	-2.6	-18.32	-
1998-04	MAM 1998	0.81	1.03	0.41	0.5	2.34	-1.4	0.8	-2.0	2.3	-
1998-05	AMJ 1998	0.07	0.4	0.31	0.45	2.84	0.3	0.8	-0.5	-15.92	-
1998-06	MJJ 1998	-0.46	-0.08	-1.0	-0.75	1.92	1.0	1.1	0.0	2.9	-
1998-07	JJA 1998	-1.05	-0.63	-1.34	-1.15	1.64	1.2	1.4	0.5	8.59	-
1998-08	JAS 1998	-1.14	-0.83	-1.33	-1.2	1.12	1.2	0.8	0.3	8.32	-
1998-09	ASO 1998	-1.24	-1.02	-1.04	-1.03	0.53	1.0	1.4	0.3	12.84	-
1998-10	SON 1998	-1.33	-1.12	-1.35	-1.34	0.79	1.1	1.2	1.2	11.09	-
1998-11	OND 1998	-1.51	-1.31	-1.31	-1.31	-0.17	1.0	1.3	1.2	11.16	-
1998-12	NDJ 1998	-1.61	-1.47	-1.69	-1.73	0.24	1.4	1.0	1.3	15.47	-

RONI and ONI are three-month means centred on the month. SOI and trade-wind indices are standardized (negative = El Niño-like). OLR in W m^{-2} (negative = more convection). Heat content in °C vs 1981–2010; WWV in 10^{14} m^3; DMI in °C; PDO standardized.

Timeline

Date	Observation	Proposed mechanism	Source	Confidence
Jul 1995 – Mar 1996	La Niña: RONI from −0.41 (JJA 1995) to a minimum of −1.02 (OND 1995); back to −0.58 by FMA 1996.	Stronger-than-normal trades build heat content in the western Pacific.	[8] [16]	High
Dec 1995 – Dec 1996	Warm water volume turns positive in December 1995 (+0.05 × 10^{14} m^3) and stays positive all of 1996	Recharge after La Niña, the precondition in recharge-oscillator theory; downwelling	[8] [14] [13] [16]	High (TAO-era data)

Date	Observation	Proposed mechanism	Source	Confidence
	(+0.26 to +0.85); heat content east of the Date Line near zero (−0.47 to +0.17 °C); heat content builds in the western Pacific for 12–18 months.	Rossby waves reflect at the western boundary as Kelvin waves only about 2 cm high in sea level.		
Jan – Apr 1996	Active MJO and a warm western Pacific, but no El Niño; westerly events in winter 1995–96 few and weak.	MJO activity alone is insufficient to start an event.	[23]	Medium–high
Apr 1996 – Jan 1997	ENSO-neutral, slightly cool: RONI −0.20 to −0.37; Niño 1+2 −0.7 to −1.6 °C (OISST).	—	[8]	High
Oct 1996 – Jun 1997	Almost continuous MJO activity: PSL OMI ≥ 1 on 23 Oct 1996 – 16 Jan 1997 (max 2.8), 25 Jan 1997 – 19 Apr 1997 (max 3.39) and 29 Apr 1997 – 13 Jun 1997 (max 3.03). Intraseasonal OLR variance over 60°E–150°E in Oct 1996 – Mar	Unusual intensification of the MJO in late 1996, possibly aided by warm western-Pacific SST.	[10] [16] [29]	High (several indices)

Date	Observation	Proposed mechanism	Source	Confidence
	1997 is the largest of 22 Oct–Mar seasons in 1979–2001.			
Mid-Oct 1996	Twin tropical cyclogenesis over the Indian Ocean during an MJO westerly burst (JTWC best tracks).	MJO westerlies straddling the equator split a super cloud cluster into two vortices.	[24]	Medium
Nov 1996 – Jan 1997	Equatorial cold SST and high SLP in the east; off-equatorial warm SST and anomalous cyclones in the far west; equatorial wind anomalies over the western Pacific switch from easterly to westerly. Trades, stronger than normal since early 1995, reverse abruptly at the end of 1996.	Off-equatorial cyclones feed equatorial westerlies.	[17] [23]	Medium–high
20–31 Dec 1996	Westerly burst: reanalysis 1000 hPa anomaly up to +11.8 m s^{-1} at 137.5°E (130°–147.5°E); TAO daily westerlies up	MJO plus cold surges from East Asia and twin cyclones; the burst excites a downwelling Kelvin wave.	[10] [9] [23] [25]	High

Date	Observation	Proposed mechanism	Source	Confidence
	to 8.7 m s^{-1} at 156°E on 21–26 Dec and again 1–3 Jan; OMI 2.66. One of two particularly intense, long-lived episodes in SSM/I winds; the MJO's own westerlies had weakened to below 2 m s^{-1} over the Maritime Continent, and the sudden intensification to over 9 m s^{-1} was confined to the Pacific.			
Jan – Feb 1997	Kelvin wave seen in TAO 20 °C isotherm depth: intraseasonal peak at 165°E on 02 Jan, at 110°W on 23 Feb (52 days, about 2.1 m s^{-1}); the sea-level signal reaches the South American coast in February. Warming near the Date Line; heat content +0.56 (Jan) and +1.00 °C (Feb).	Downwelling Kelvin wave deepens the eastern thermocline on top of the recharged state.	[9] [23] [17] [8]	High

Date	Observation	Proposed mechanism	Source	Confidence
27 Feb – 19 Mar 1997	The strongest event of the winter (stress up to 0.2 N m^{-2}, fetch over 30° of longitude, about a month long). TAO westerlies up to 9.9 m s^{-1} at 147°E (27 Feb – 6 Mar) and 8.6 m s^{-1} at 165°E, reaching 180° (11–16 Mar); reanalysis burst 3 Mar 1997 – 19 Mar 1997, peak +10.3 m s^{-1} at 150°E, spanning 137.5°E–177.5°W; OMI 2.75. Western-Pacific 850 hPa trade index −3.1 for March.	Kelvin wave plus rapid eastward advection of the warm pool's edge; the atmosphere's response to these SST changes favours further westerlies.	[27] [9] [10] [8]	High
Mar 1997	Model response to the March event: far-western cooling of about 0.8 °C, the warm pool's eastern edge displaced about 2000 km in a month, warming of about 0.5 °C along the Kelvin-wave	Kelvin-wave and nonlinear advection at the warm-pool edge.	[26] [27] [28]	High (ocean and atmosphere model experiments for this ever

Date	Observation	Proposed mechanism	Source	Confidence
	path; surface jets at the warm-pool edge three times the linear response.			
Mar – Apr 1997	Far-eastern warming (Niño 1+2 −0.45 °C in Feb, +0.23 in Mar, +0.80 in Apr). Late March – early April: central-Pacific warming as the Kelvin wave forced by the February MJO passes, mainly by zonal advection. TAO shows the next Kelvin wave at 165°E on 16 Mar and at 110°W on 23 Apr (38 days, about 2.88 m s^{-1}).	Coastal arrival of the earlier wave; Kelvin-wave-induced advection with a lasting change in winds, currents and SST gradient.	[17] [29] [9] [8]	High
10–21 Apr 1997	Westerly burst: reanalysis peak +10.8 m s^{-1} at 155°E; TAO westerlies up to 11.0 m s^{-1} at 165°E (13–20 Apr). The OLR-based OMI shows an active MJO (2.01) while the	Recharged reservoir at its maximum as coupling begins; burst during an eastward-extended warm pool.	[10] [9] [8] [7]	High (event); medium (MJO link)

Date	Observation	Proposed mechanism	Source	Confidence
	circulation-weighted index is weak (0.9). Warm water volume peaks at +3.12 × 10^{14} m^3; heat content +2.17 °C.			
1 Jun 1997	Adjoint ocean-model decomposition of the modelled Niño 3 anomaly: initial thermal state about 1.1 °C and westerly anomalies about 1.0 °C (about 40% and 50% relative to an average year); the early-March, late-March and early-April wind peaks contribute about 0.6, 0.3 and 0.5 °C, the December event about 0.4 °C over a negative baseline. Compared with 1996: about 30% from the stronger build-up and about 90% from the westerly events.	Quantitative split between precondition and wind triggers.	[25]	High for the model result; one model

Date	Observation	Proposed mechanism	Source	Confidence
May – Jun 1997	Coupling established: RONI AMJ +0.83, MJJ +1.20; SOI −1.3 and −1.4; central-Pacific OLR −25.7 and −26.8 W m^{-2}; Niño 1+2 +1.83 and +2.79 °C. More bursts (19 May 1997 to 12 Jun 1997; TAO 10.2 m s^{-1} at 165°E on 22–26 May) and twin tropical cyclogenesis over the western Pacific in late May. The burst series that began in December ends in June.	Bjerknes feedback takes over once the central and eastern warm regions join.	[8] [18] [9] [10] [24]	High
Jun – Dec 1997	Eastern equatorial SST sets a new monthly record every month from June to December; the cold tongue fails to form in boreal summer and fall; westerlies and deep convection only over water warmer than about 29 °C.	Coupled growth with the warm pool shifted east.	[15] [16]	High

Date	Observation	Proposed mechanism	Source	Confidence
Jul – Oct 1997	RONI JJA +1.50 → SON +2.22; central-Pacific trade index −4.1 and OLR −49.8 W m^{-2} in October; much of the western warm water carried east by anomalous surface currents.	Coupled growth, with zonal advection important in the central Pacific.	[8] [18] [54]	High
Jun 1997 – Feb 1998	Indian Ocean: warm SST off East Africa from June 1997 (peak above +2 °C in Feb 1998); cool SST in the east from July (anomaly of about 2 °C at its November 1997 maximum); equatorial winds reverse to easterly, anomalies beyond −5 m s^{-1} in December; east–west sea-level difference above 30 cm Nov 1997 – May 1998. CPC dipole index +0.55 °C (Jul) to +1.39 °C (Nov).	Disputed: forced by ENSO through the Walker circulation and monsoon, or an internal Indian Ocean mode.	[47] [51] [49] [50] [8]	High (observation); mechanism disputed

Date	Observation	Proposed mechanism	Source	Confidence
Late 1997	Eastern thermocline more than 90 m deeper than normal; TAO 20 °C isotherm at 110°W about 130 m in Oct–Dec 1997 against 29–46 m in Dec 1996 – Jan 1997.	Cumulative effect of the downwelling Kelvin waves and coupled growth.	[16] [9]	High
Nov 1997	Thermocline shoaling begins in the central Pacific and propagates east; near-Date-Line westerly anomalies shift south of the equator; warm water volume +0.66 and negative in December (-0.77×10^{14} m^3).	Seasonal southward shift of the warmest water and its winds forces eastern thermocline shoaling; discharge begins.	[57] [58] [55] [59] [8]	High
NDJ 1997	Peak: RONI +2.28 (ONI +2.37). Cold-tongue anomalies averaged nearly 4 °C in December, with maxima above 5 °C.	Mature coupled state; discharge under way.	[8] [15] [16]	High
Dec 1997 – Apr 1998	Eastern SST stays warm despite strong subsurface cooling; easterly	Upwelling Kelvin waves from western easterlies (and, secondarily,	[57] [58] [16] [18] [9]	High

Date	Observation	Proposed mechanism	Source	Confidence
	anomalies in the far western Pacific; westerlies weaken and migrate east with the 29 °C water; easterlies vanish from the eastern equatorial Pacific in February 1998. TAO 20 °C isotherm at 140°W rises from about 135 m (Nov 1997) to about 62 m (May 1998).	boundary reflection) and Rossby waves from eastern westerlies; maximum shoaling where their paths cross, 125°W–170°W.		
Mid-May 1998	Trades return to near normal in the east; SST at 0°, 125°W falls 8 °C in 30 days, more than ten times the normal rate; eastern equatorial SST falls by more than 4 °C within two weeks. An MJO is active (OMI ≥ 1 from 30 Apr 1998 to 31 May 1998, max 2.24).	Thermocline surfaces once easterlies return; an MJO is implicated in the timing; the cause of the trade-wind return was not known at the time.	[15] [57] [56] [10]	High (observation); medium (MJO trigger)
Jun – Aug 1998	La Niña: RONI MJJ −0.46, JJA −1.05; the largest equatorial	Upwelling resumes over a discharged basin.	[8] [22]	High

Date	Observation	Proposed mechanism	Source	Confidence
	phytoplankton bloom then observed.			

Confidence refers to the observation and its link to the stated mechanism for this event, not to general ENSO theory.

Evidence table

Factor	Observed conditions and timing	Proposed mechanism	Role in development	Evidence type	Confidence	Key sourc
Warm-water recharge after the 1995–96 La Niña	WWV positive from Dec 1995, +3.12 × 10^{14} m^3 in Apr 1997; adjoint model credits about 40% of the 1 June 1997 Niño 3 anomaly to the initial state, concentrated in the deep western warm layer.	A charged reservoir lets Kelvin waves raise eastern SST and sustain growth.	Precondition	A−: TAO-era observation + adjoint decomposition for this event	High	[8] [1 [25] [16]
Westerly wind events, Dec 1996 – Jun 1997	10 reanalysis bursts; 16 in calendar 1997, the most of any year in 1979–2001 (mean 4.2; 1982: 13; 1995: 0; 1996: 3); confirmed by TAO mooring winds. Wind stress ≈ 50% of the 1 June anomaly and ≈ 90% of the	Downwelling Kelvin waves (two traced in TAO: about 2.1 and 2.9 m s^{-1}) and eastward displacement of the warm-pool edge.	Trigger (and early amplifier)	A: event observations (TAO, satellite, reanalysis) + adjoint decomposition + targeted experiments	High	[25] [23] [[10] [26] [18]

Factor	Observed conditions and timing	Proposed mechanism	Role in development	Evidence type	Confidence	Key sourc
	difference from 1996.					
March 1997 event: atmospheric response	The model SST response to the March event weakens the central-eastern trades and shifts convection east, promoting further frequent and intense events.	Burst-forced SST changes → more westerlies.	Amplifier of the trigger	A−: ocean + atmosphere GCM experiments for this event	Medium–high	[27] [28]
MJO, Oct 1996 – Jun 1997	OMI ≥ 1 almost continuously; the largest intraseasonal OLR variance over 60°E–150°E of 22 seasons; bursts on 25 Dec, 13 Mar and 28 Mar during OMI 2.7–3.3. MJO-forced Kelvin waves 15–20 cm in sea level against about 2 cm from boundary reflection. But an active MJO in Jan–Apr 1996 produced no El Niño, and burst intensification	MJO organises bursts; rectified MJO stresses add warming; Kelvin waves advect warm water.	Trigger timing / amplifier	B: event-specific observations and wave model; one intermediate-model hindcast (+50% amplitude)	Medium–high	[10] [[29] [23] [30]

Factor	Observed conditions and timing	Proposed mechanism	Role in development	Evidence type	Confidence	Key sourc
	exceeded the MJO's own signal.					
Cold surges and twin cyclones	Northerly cold surges from East Asia linked to equatorial twin circulations that enhanced the late-Dec 1996 and late-Feb 1997 bursts (SSM/I winds). JTWC best-track twin cyclogenesis is found in mid-Oct 1996 (Indian Ocean) and late May 1997 (western Pacific), not at the two main bursts.	Extratropical forcing amplifies bursts.	Amplifier of the trigger (extratropical pathway)	B: two event-specific studies, partly differing in what counts as a twin cyclone	Medium	[23] [24]
Off-equatorial western-Pacific cyclones (Nov 1996 – Jan 1997)	Off-equatorial warm SST and cyclones accompany the switch to equatorial westerlies.	Off-equatorial circulation forces equatorial westerlies.	Trigger component (proposed)	B/C: one descriptive study	Medium	[17]
Western-boundary Rossby wave reflection	Reflected downwelling Kelvin waves in 1995–96 only about 2 cm in sea level, dwarfed by MJO-forced waves;	Delayed-oscillator reflection.	Secondary at onset and termination	B: wave model + altimetry for this event	Medium–high	[16] [[19] [

Factor	Observed conditions and timing	Proposed mechanism	Role in development	Evidence type	Confidence	Key sourc
	upwelling reflections (4–6 cm) contribute to the slow shoaling of early 1998.					
Bjerknes coupled feedback	From May–June 1997 SOI, trades, OLR and SST strengthen together to an October–December peak; record eastern SST every month June–December.	Warming → weaker trades → deeper eastern thermocline → more warming.	Amplifier / feedback	B: event observations + general theory; heat budgets show zonal advection important	High	[8] [1 [18] [54]
State dependence of bursts	Bursts are more likely when the warm pool extends east; bursts continued after onset (seven more in the reanalysis from August to December 1997, mostly east of 160°E).	Bursts partly a response to ENSO, effectively increasing coupling.	Feedback (contested)	C: multi-event statistics and models, not isolated for 1997	Medium	[37] [38] [36] [34] [35] [10]
Pacific meridional mode / seasonal footprinting	Most El Niños of the past four decades preceded by spring warming and south-westerlies near the ITCZ.	Extratropical winter variability → subtropical SST → equatorial winds.	Possible precondition	C: composite and model studies; 1997 not isolated in the sources read	Low	[42] [43] [44]

Factor	Observed conditions and timing	Proposed mechanism	Role in development	Evidence type	Confidence	Key sourc
Pacific decadal background	PDO positive through 1997 (+1.6 to +2.6 in May–Jun); shallow meridional overturning at a 1992–98 low.	Weaker overturning → warmer tropical background; background alters the response to bursts.	Background (possible precondition)	C: decadal observations and a simple model; not event attribution	Low	[8] [4 [46] [39]
Indian Ocean Dipole 1997	Western warming from June and eastern cooling from July 1997, after Pacific coupling; peak dipole index +1.39 °C (Nov).	ENSO-forced (divergent easterlies over the equatorial Indian Ocean in summer 1997; reversed Walker circulation) versus an internal Indian Ocean mode (16 gradient reversals in 1950–98, only three in El Niño years).	Consequence for onset; origin disputed; feedback on the Pacific untested here	B: event-specific observational and model studies that disagree	Medium	[51] [50] [49] [47] [48] [52]
Tropical North Atlantic	TNA near or below zero in early 1997 (−0.37 °C in Mar), warm in 1998 (+0.51 °C in May).	Cool NTA in spring can favour El Niño; warming in 1998 is the usual lagged response.	Untested trigger; later consequence	C/D: index only for this event	Low	[8] [5
Seasonal cycle in termination	Southward shift of westerly anomalies from Nov 1997; thermocline shoaling from	Seasonal migration of the warmest SST shifts winds south, shoaling the eastern thermocline.	Termination mechanism	A: event observations + AGCM and OGCM experiments	High	[55] [57] [58] [59]

Factor	Observed conditions and timing	Proposed mechanism	Role in development	Evidence type	Confidence	Key sourc
	the central Pacific; abrupt SST drop in May 1998.					
MJO at termination (May 1998)	An MJO active from 30 Apr 1998 to 31 May 1998 (OMI max 2.24) as the trades returned in mid-May.	Easterly phase helps restore the trades over a shallow thermocline.	Timing of termination	B: one event-specific study (title only read) + index	Medium	[56] [10] [
Volcanic forcing	No eruption of VEI ≥ 4 anywhere in 1995–97 in the GVP catalogue; the most recent tropical VEI 4 was Rabaul (4°S) in September 1994.	—	Not a candidate	Catalogue check	High	[11]
Background warming	RONI peak +2.28 vs ONI +2.37: the tropical-mean adjustment changes the peak by 0.09 °C.	Index statement, not attribution.	Not attributed	D for event attribution; projections only	—	[2] [6 [64]

Evidence grades: A = event-specific observation plus a model or heat-budget study isolating the effect; B = event-specific observation with a plausible mechanism and supporting studies; C = correlation, composite or general theory only; D = speculative, disputed, or not yet tested for this event.

Westerly wind events and Kelvin waves, December 1996 – June 1997

Reanalysis bursts on Paper 1's definitions (1000 hPa anomaly ≥ 5 m s^{-1}, ≥ 3 days, ≥ 10° of longitude, 5°S–5°N, 120°E–150°W), with the PSL OLR MJO index (OMI) and the circulation-weighted index at the burst peak [10].

Start	End	Peak	Peak anomaly (m s^{-1})	Peak lon	Extent	OMI	Circulation index
1996-12-20	1996-12-31	1996-12-25	11.8	137.5°E	130°E–147.5°E	2.66	3.85
1997-02-27	1997-03-04	1997-02-28	7.0	132.5°E	125°E–155°E	1.77	2.81
1997-03-03	1997-03-19	1997-03-13	10.3	150°E	137.5°E–177.5°W	2.75	4.39
1997-03-25	1997-03-29	1997-03-28	7.1	165°E	157.5°E–175°W	3.31	3.71
1997-04-10	1997-04-21	1997-04-13	10.8	155°E	137.5°E–167.5°E	2.01	0.9
1997-05-02	1997-05-08	1997-05-05	7.3	167.5°E	160°E–172.5°E	1.95	0.12
1997-05-19	1997-05-24	1997-05-20	7.3	140°E	135°E–150°E	2.48	2.17
1997-05-21	1997-05-27	1997-05-26	7.7	162.5°E	152.5°E–180°E	2.90	2.95
1997-05-30	1997-06-12	1997-06-04	9.3	167.5°W	162.5°E–150°W	1.95	2.87
1997-06-21	1997-06-23	1997-06-22	6.2	175°E	170°E–180°E	0.48	0.89

Directly observed TAO episodes (daily westerly ≥ 5 m s^{-1} on ≥ 2 days at 147°E, 156°E, 165°E or 180°): 19 Nov 1996 – 21 Nov 1996 (6.7 m s^{-1} at 156°E); 21 Dec 1996 – 22 Dec 1996 (7.5 m s^{-1} at 156°E); 24 Dec 1996 – 26 Dec 1996 (8.7 m s^{-1} at 156°E); 1 Jan 1997 – 3 Jan 1997 (7.2 m s^{-1} at 156°E); 14 Jan 1997 – 15 Jan 1997 (6.7 m s^{-1} at 156°E); 27 Feb 1997 – 6 Mar 1997 (9.9 m s^{-1} at 147°E); 11 Mar 1997 – 16 Mar 1997 (8.6 m s^{-1} at 165°E); 27 Mar 1997 – 30 Mar 1997 (5.5 m s^{-1} at 165°E); 13 Apr 1997 – 20 Apr 1997 (11.0 m s^{-1} at 165°E); 22 May 1997 – 26 May 1997 (10.2 m s^{-1} at 165°E); 1 Jun 1997 – 3 Jun 1997 (7.7 m s^{-1} at 165°E); 7 Jun 1997 – 12 Jun 1997 (7.3 m s^{-1} at 165°E). TAO wind records are incomplete at 147°E (37% of days) and 156°E (64%).

MJO contribution (assessment)

Answer. Strong evidence that the MJO set the timing and much of the amplitude of the onset bursts, through Kelvin waves an order of magnitude larger than boundary reflections [16] [29]; but the MJO was not sufficient on its own — an active MJO in early 1996 produced no El Niño, and the bursts that mattered were intensified locally by cold surges and twin circulations [23]. Bergman et al. conclude that the MJO is relevant to the timing and initial growth of El Niño rather than responsible for the event itself [29]. One intermediate coupled model initialised on 1 January 1997 becomes about 50% stronger when rectified MJO winds are included [30]. Statistically, MJO events warm the waveguide only when they contain a westerly wind event [35].

Compared with 1982. The 1996–97 season had 137 days with the circulation-weighted index at or above 1 against 77 in 1981–82, and the bursts in calendar 1997 outnumber those of 1982 (16 vs 13) on identical definitions [10]. In the mature phase (Oct 1997 – Mar 1998) intraseasonal OLR variance over 60°E–150°E fell to 77.8 $W^2\ m^{-4}$ (22-season mean 233), consistent with the weak eastward MJO propagation reported for the mature stage [41].

Rainfall and tropospheric temperature (GPCP, NOAA STAR)

Rainfall in the usual El Niño impact regions, from GPCP v2.3 against 1991–2020. In 1997–98, 7 of the 8 regions had the usual El Niño sign. The most extreme seasons (top three since 1979) were: Central equatorial Pacific (SON, 394% of normal, 1st wettest); Coastal Peru and Ecuador (Dec–Apr, 269% of normal, 1st wettest); Indonesia (Jul–Oct, 50% of normal, 2nd driest); Equatorial East Africa (Oct–Dec, 349% of normal, 1st wettest); California (Nov–Mar, 180% of normal, 2nd wettest). The same comparison across all four events is in Paper 4.

The tropical lower troposphere (NOAA STAR TLT, 20°S–20°N) warmed to +1.12 °C above 1991–2020 by Feb 1998, the usual lag of a few months behind the sea surface.

Region (season)	Usual El Niño signal	1997–98
Central equatorial Pacific (5°S–5°N, 170°E–140°W) (SON)	wetter	394% 1st wettest of 47
Coastal Peru and Ecuador (12.5°S–2.5°N, 85°–77.5°W) (Dec–Apr)	wetter	269% 1st wettest of 47
Indonesia (10°S–5°N, 95°–145°E) (Jul–Oct)	drier	50% 2nd driest of 47
Eastern Australia (40°–10°S, 140°–155°E) (Jun–Nov)	drier	77% 8th driest of 47
India (8°–30°N, 70°–90°E) (Jun–Sep)	drier	101% 28th driest of 47
Equatorial East Africa (5°S–5°N, 35°–50°E) (Oct–Dec)	wetter	349% 1st wettest of 47
Southern Africa (35°–15°S, 15°–40°E) (Dec–Mar)	drier	94% 22nd driest of 47
California (32.5°–42.5°N, 125°–115°W) (Nov–Mar)	wetter	180% 2nd wettest of 47

GPCP v2.3 monthly precipitation (NOAA PSL), box means over land and ocean on the 2.5° grid; percentage of the 1991–2020 mean for the same months; rank among the 47 seasons 1979–2025 (wettest for regions that usually turn wet in El Niño, driest for those that usually turn dry). Data to Jun 2026. Seasons are the usual El Niño impact seasons, counted from the onset year (Dec–Apr = December of the onset year to April of the next).

Stage 5 verification log (23 October 2026)

Data. Every number in the text that comes from the monthly table, the TAO analysis or the reanalysis and MJO diagnostics was recomputed from the source files by `tools/papers/verify_1997.py`: 75 checks, all passing after the correction below.

Diagnostic values in the text are inserted by the build script from `data/papers/1997/diag_1997.json`, not typed by hand.

Literature. Claims were re-read against the full texts held in the private library: McPhaden (1999), McPhaden and Yu (1999), Yu and Rienecker (1998), van Oldenborgh (preprint of 2000), Lengaigne et al. (2003), Bergman et al. (2001), Webster et al. (1999), Ueda and Matsumoto (2000), Zhu et al. (2003), Barnston et al. (1999, NCEP), Landsea and Knaff (2000), Kessler and Kleeman (2000), Vecchi and Harrison (2006), Vecchi (2006), Kutsuwada and McPhaden (2002) and Picaut et al. (2002). All other claims rest on the abstracts recorded in the reference list. **Not read beyond the bibliographic record:** Saji et al. (1999), Takayabu et al. (1999), Boulanger and Menkes (1999), Ham et al. (2013), McPhaden and Zhang (2002); each is cited only for what its title or a held paper states.

Corrections made.

- Start of the La Niña phase: “−0.41 (JAS 1995)” → −0.41 (JJA 1995); JAS 1995 was −0.57.
- Eastern Indian Ocean cooling: “below −2 °C in Nov 1997” → an anomaly of about 2 °C at its November maximum (the sign is garbled in the extracted text of Webster et al.; magnitude kept, wording neutral).

Checked and unchanged. The adjoint numbers (0.6, 0.3, 0.5 and 0.4 °C; 40%/50%; 30%/90%) match the preprint and the journal abstract; the 8 °C in 30 days at 0°, 125°W, the June–December monthly records and the >90 m thermocline depression match McPhaden (1999); the 15–20 cm against about 2 cm Kelvin-wave amplitudes match McPhaden and Yu (1999); the forecast statements match Barnston et al. (1999, both papers) and Landsea and Knaff (2000).

Grades. No evidence grade was changed in this pass. **Open items:** island sea-level records (UHSLC server unreachable from the automated fetch); an ERA5 re-test of the burst list, as for 1982.

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Every DOI was checked against the Crossref registry; summaries rely on the published abstract unless noted. Full texts behind publisher paywalls were not read and are marked.

The four papers

Data snapshot: 24 October 2026. Independent, non-commercial analysis built from public sources (see About & method). Analysis prepared with Claude (Anthropic).